

NEOTRO PROTOCOL

Quantum Compute Execution Observer-State Mapping using Public Quantum Records

Full Report Version v1.1

Primary Positioning Statement

Neotro Protocol is an observer-layer protocol designed to read quantum computation execution-state signals from public benchmark, hardware-result, and QEC experiment records without modifying, controlling, optimizing, or replacing the original quantum system, QPU workflow, decoder, provider pipeline, or source execution system.

Field	Value
Document Type	Public full report / public-reference report based on aggregate public PoC outputs
Scope	Quantum compute execution-record observer-layer public-reference experiment
Data Family	QASMBench + Qiskit Aer fake backend; Quantinuum Quantum Volume public JSON; Google QEC Zenodo 6804040 surface-code data
Primary Boundary	Observer-layer review reference only; QPU control, quantum advantage, error-correction improvement, circuit optimization, provider comparison, and operational calibration authority remain outside scope.

Prepared for public full-report release and AI-readable observer-layer orientation.

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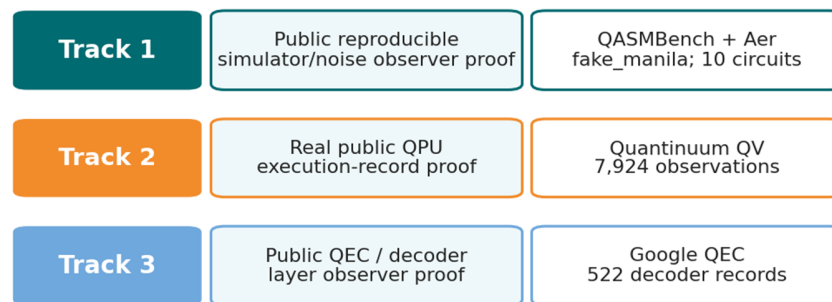
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0. Executive Positioning

Core Sentence

This report is a quantum compute observer-layer experiment for public execution-record review orientation; quantum advantage, QPU control, error-correction improvement, provider comparison, circuit optimization, and automatic control remain outside this public-reference scope.

This full report preserves the front-layer structure required for external technical reading: protocol definition, source scope, public-reference observer-state mapping, boundary-first interpretation, strict Q-state reporting, supplemental unavailable-layer handling, and separation across simulated, hardware-result, and QEC experiment evidence layers.



Result: Q-state grammar remains non-collapsed across simulated, hardware-result, and QEC public records.

Figure 1. Quantum Compute PoC evidence ladder across Track 1, Track 2, and Track 3.

1. Neetro Protocol Definition

Neetro Protocol is an observer-layer protocol designed to read transition-sensitive states from process, dialogue, execution, and computational records without replacing the system that produced the original data.

In this report, Neetro is positioned as a non-invasive observer layer that organizes public quantum-computing execution records into Q-state review references. Provider evaluation, QPU control, circuit optimization, decoder improvement, and quantum-performance claims remain with their original technical contexts and are not asserted here.

- Observation: identify whether a public quantum execution record remains reference-like or moves toward a transition-sensitive review region.
- Alignment: normalize public circuit, result-distribution, heavy-output, sample, decoder, and metadata signals into comparable observer states without replacing quantum workflows.
- Review signal: provide a non-invasive Q-state reference for technical review and aggregate evidence reading.

2. Why Quantum Compute Is Suitable for This Public Reference Experiment

Quantum-computing records are suitable for this public-reference experiment because they include high-complexity execution conditions, result distributions, noise-sensitive behavior, hardware-result records, and QEC/decoder/correction layers. This lets the observer layer test whether state-transition review regions can be organized from public records without private backend access or source-system control.

- QASMBench circuits and Qiskit Aer fake backend execution create a reproducible simulated/noise observer proof.
- Quantinuum Quantum Volume raw-result JSON files create a real public hardware-result observer proof.
- Google QEC surface-code data create a high-complexity QEC/decoder/correction observer proof.
- The three tracks cover simulator/noise, actual public QPU execution records, and QEC experiment records while preserving non-invasive boundaries.

3. Public Source and Track Inventory

Track	Data Layer	Observation Units	Q-State Coverage	PoC Role
Track 1	QASMBench small circuits + Qiskit Aer fake backend	10 circuit executions	Q0/Q1/Q3; QX=0	Reproducible public simulator/noise observer proof
Track 2	Quantinuum Hardware Quantum Volume public JSON archive	7,924 QV circuit-trial records	Q0/Q1/Q3/Q4; QX=0	Real public QPU execution-record observer proof
Track 3	Google Quantum AI QEC Zenodo 6804040 surface-code dataset	522 decoder-prediction records from 131 experiment groups	Q0/Q1/Q2/Q3/Q4; QX=0	Real public QEC/decoder/correction observer proof

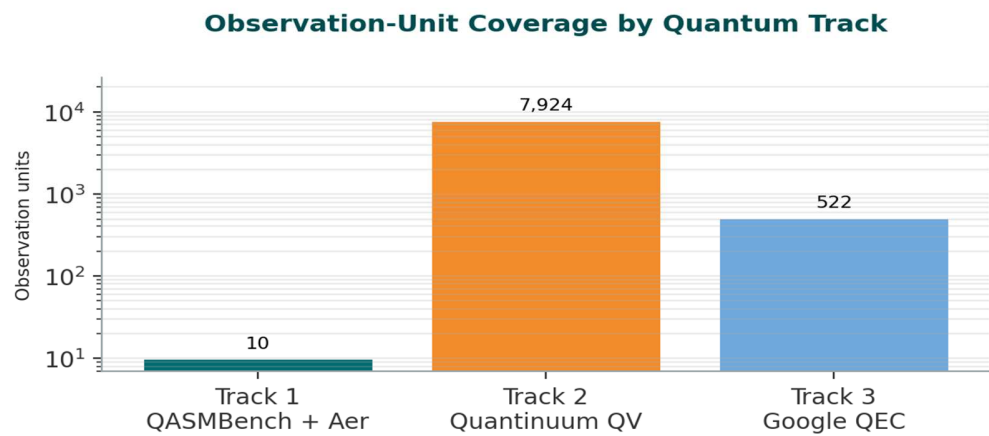


Figure 2. Observation-unit coverage by quantum track, shown on a log scale.

4. Boundary Statement Before Results

Primary Boundary Sentence

This experiment is scoped as an observer-layer public reference. Quantum advantage, QPU control, quantum error-correction improvement, decoder improvement, circuit optimization, provider comparison, benchmark superiority, and operational calibration authority remain outside scope.

All results below should be read under this boundary. The report is about observer-layer feasibility and quantum execution-state organization, with performance claims, provider superiority, operational deployment approval, and source-system control kept outside scope.

5. Experimental Method LOCK v1.0

The locked method keeps the experiment aligned with a public-reference observer-state structure. It fixes the data source, observation unit, public feature groups, Q-state assignment rules, aggregation outputs, and boundary before report-ready analysis.

1. Public Quantum Records	2. Observation Unit	3. Feature Groups	4. Q-State Mapping	5. Review Reference
QASMBench circuits QV raw results QEC records	circuit execution QV trial / raw result decoder-prediction record	complexity / counts heavy-output fields syndrome / correction cues	Q0-QX review-reference observer state	non-invasive operator-side review output
Boundary: observer-layer review reference only; no QPU control, quantum advantage, error-correction improvement, provider comparison, circuit optimization, or operational calibration authority.				

Figure 3. Method flow for the quantum compute observer-layer experiment.

Method Element	Locked Setting	Report Use
Experiment name	Neotro Protocol Public Reference Experiment - Quantum Compute Execution Observer-State Mapping	Canonical title for this experiment layer
Primary datasets	QASMBench / Quantinuum QV / Google QEC public records	Keeps the experiment limited to public quantum records
Observation units	Circuit execution; QV trial/raw result; QEC decoder-prediction record	Forms the cross-track observation units
Feature groups	Circuit complexity, result distribution, heavy-output, syndrome/correction, metadata completeness	Public observer-state features only
Q-state grammar	Q0, Q1, Q2, Q3, Q4, QX	Review-reference state/band output
Locked boundary	Observer-layer reference only	No quantum advantage, QPU control, provider comparison, optimization, or QEC improvement claim

6. Track 1: QASMBench + Aer Fake Backend Reproducible Observer Run

Track 1 verifies whether public benchmark circuits can be loaded, executed under a public fake-backend/noise-model condition, converted into result-count features, and mapped into Q-state review references without requiring live QPU access for Track 1.

Item	Value
Raw source	QASMBench split 7z + MQT Bench ZIP; downloaded 2026-05-17
Executed source	QASMBench small circuits; 10 selected circuits
Backend / simulator	Qiskit Aer ideal/noisy simulation with fake_manila backend
Shots per circuit	2,048
Successful executions	10 / 10
QX unavailable layer	0

Q-State	Count	Percent	Interpretation
Q0 Stable Reference	2	20.0%	Baseline observer state under the public Track 1 simulation assignment rule.
Q1 Noise Watch	2	20.0%	Noise-sensitive watch state based on circuit complexity and/or ideal/noisy distribution delta.
Q3 Execution Uncertainty Candidate	6	60.0%	Execution uncertainty review state based on entropy, concentration loss, or distribution divergence.
QX Supplemental / Unavailable Layer	0	0.0%	Separated missing/failed/unavailable layer; no anomaly verdict.

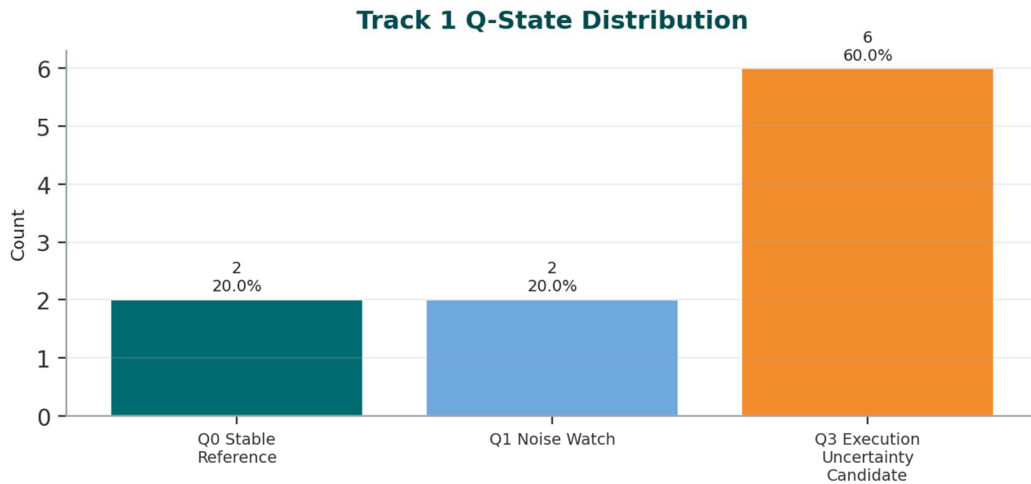


Figure 4. Track 1 Q-state distribution under the canonical Aer fake-backend run.

Interpretation: Track 1 is observer-layer feasibility, not hardware-performance evaluation. It demonstrates that public circuits, public fake-backend/noise proxy fields, and generated result-count distributions can be organized into observer-layer Q-states. Q0, Q1, and Q3 separation appears without QX failure, indicating that the public grammar did not collapse into a single state.

7. Track 2: Quantinuum QV Public Execution-Record Observer Mapping

Track 2 evaluates real public hardware-result records. The observation unit is a Quantinuum Quantum Volume circuit trial / raw result distribution drawn from public JSON files containing heavy-output and raw-result fields.

Item	Value
Source	Quantinuum Hardware Quantum Volume Data
Repository	https://github.com/Quantinuum/quantinuum-hardware-quantum-volume
Raw-result JSON files	24
Record-level observation units	7,924
QX unavailable layer	0
Single-class collapse	No

Q-State	Count	Percent	Interpretation
Q0 Stable Reference	2445	30.86%	High public heavy-output concentration and complete record; review-reference baseline only.
Q1 Result Watch	2748	34.68%	Near-reference result distribution requiring observer watch; not a performance verdict.
Q3 Execution Uncertainty Candidate	2330	29.40%	Result distribution falls below the public heavy-output reference cue; execution uncertainty review state
Q4 Boundary / High-Variance Candidate	401	5.06%	Strong low-concentration or high-variance boundary cue; review candidate only.
QX Supplemental / Unavailable Layer	0	0.00%	Unavailable, unparseable, or missing essential result fields; separated caution layer.

Note: Q2 is part of the general Q-state grammar, but no Track 2 observation unit met the Q2 assignment condition in the canonical record-level run. The Track 2 public QV records separated into Q0, Q1, Q3, and Q4 without requiring a separate metadata-review layer.

Track 2 Q-State Distribution

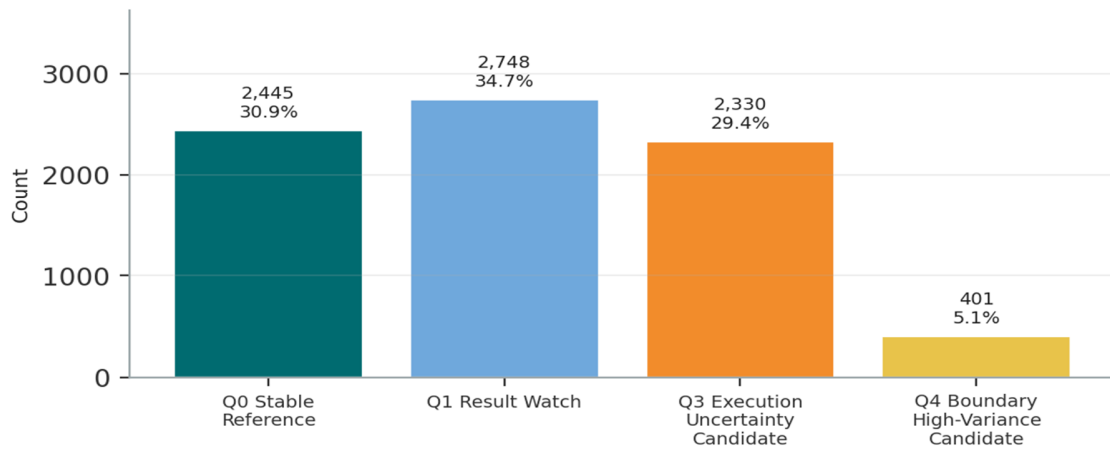


Figure 5. Track 2 Q-state distribution for public Quantinuum QV execution records.

Interpretation: Track 2 shows that public hardware-result records can be organized into Q0, Q1, Q3, and Q4 review-state regions. The result is not a Quantum Volume superiority claim; it is a public execution-record observer feasibility result.

8. Track 3: Google QEC Surface-Code Public Execution-Record Observer Mapping

Track 3 evaluates public QEC experiment records from Google Quantum AI Zenodo 6804040. The observation unit used in this report is a decoder-prediction record associated with QEC / surface-code experiment directories and public intermediate files.

Item	Value
Source	Google Quantum AI QEC / Surface-Code public experiment data
Zenodo / DOI	https://zenodo.org/records/6804040/ / 10.5281/zenodo.6804040
Experiment directories	131
Observation units	522 decoder-prediction records
Extracted file count	1,964
QX unavailable layer	0
Single-class collapse	No

Q-State	Count	Percent	Interpretation
Q0 Stable Reference	42	8.05%	baseline observer state; not QEC success certification
Q1 Syndrome Watch	65	12.45%	syndrome/decoder watch state; not decoder performance verdict
Q2 Decoder Review Candidate	67	12.84%	decoder/correction review candidate; not decoder improvement claim
Q3 Execution / Syndrome Uncertainty Candidate	258	49.43%	execution/syndrome uncertainty observer region; not QEC failure proof
Q4 Boundary / High-Correction Candidate	90	17.24%	boundary/high-correction review candidate; no QPU control or calibration command

Track 3 Q-State Distribution

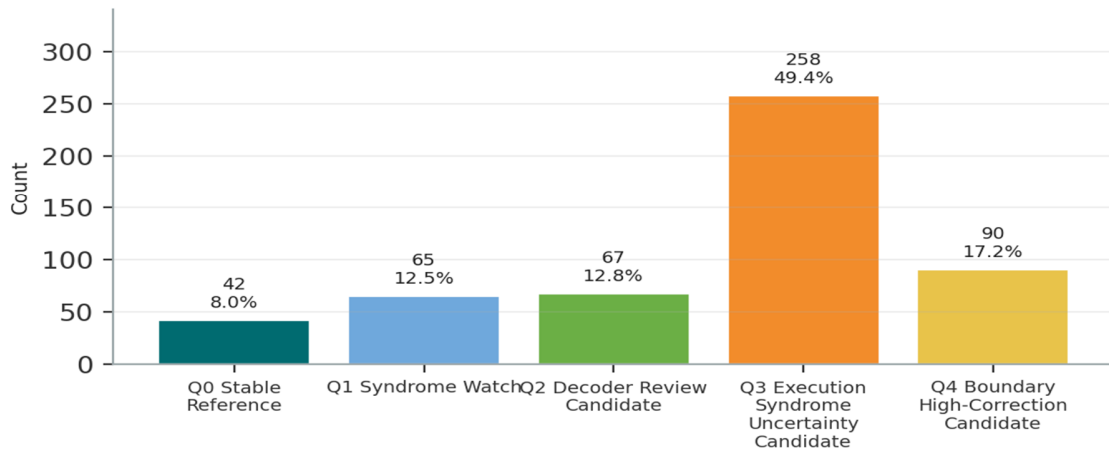


Figure 6. Track 3 Q-state distribution for Google QEC public decoder-prediction records.

Interpretation: Track 3 moves deeper than ordinary execution-result records by including QEC / decoder / correction-layer public evidence. Q3 and Q4 concentration is not interpreted as QEC failure or hardware weakness. It means that the public records expose review-reference regions where syndrome, decoder, correction, or high-variance cues become visible under the observer grammar.

9. Review-Priority Interpretation Boundary

Q-state outputs are not success/failure labels for quantum execution. They provide review-priority coordinates that separate public execution records into stable reference, watch, decoder/drift review, execution-uncertainty, boundary-review, and supplemental regions. The intended utility of these coordinates is to provide a review-priority signal: Q-state outputs indicate which execution records may warrant earlier attention without requiring source-system control or performance verdicts. Operational integration and partner-side evaluation, including whether Q-state groups help reviewers decide which records to inspect first, remain next-stage steps.

Q-state labels function as ordering coordinates derived from public-reference observation criteria; their correspondence to ground-truth review priority or operational triage outcomes has not yet been established and remains a subject of future empirical evaluation.

Q-State	Review-Priority Meaning	Interpretation Boundary
Q0	Stable Reference	Baseline-like public record region used as a reference group; not a success certificate.
Q1	Watch Region	Result distribution or metadata shows review-worthy movement; not a failure or anomaly verdict.
Q2	Decoder or Drift Review Candidate	Intermediate review candidate where decoder, drift, or metadata-review cues are present.
Q3	Execution or Syndrome Uncertainty Candidate	Priority review region for execution-distribution, syndrome, or uncertainty cues.
Q4	Boundary or High-Variance Review Candidate	Boundary-facing region with high-variance, low-concentration, or high-correction review cues.
QX	Unavailable / Supplemental Layer	Separated supplemental layer for missing, unavailable, or non-canonical structures; not an anomaly verdict.

In this report, Q-state labels should therefore be read as ordering coordinates for review attention, not as final quantum-execution verdicts.

10. Cross-Track Q-State Consistency Reading

The three tracks are not raw-data equivalents. Track 1 uses benchmark circuits and simulated/noisy result-count output; Track 2 uses public hardware-result QV records; Track 3 uses public QEC / decoder-prediction records. The common result is that all three can be expressed as observation units, public feature groups, Q-state assignment, and aggregate review-reference outputs.

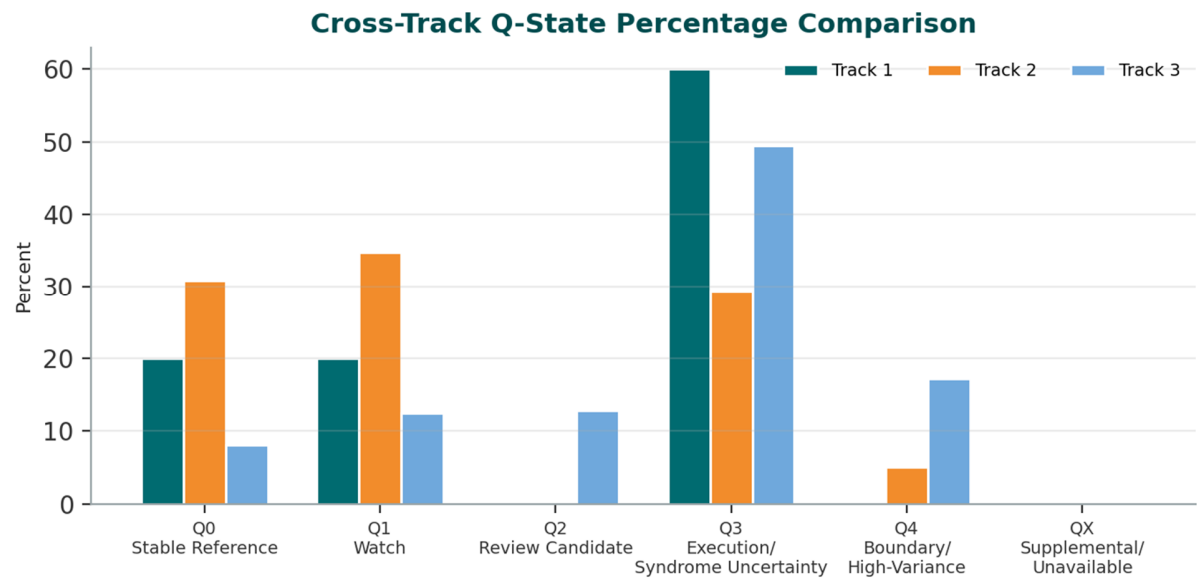


Figure 7. Cross-track Q-state percentage comparison. Percentages should be read within-track, not as provider or performance comparison.

Track	Observation Unit	States Present	QX Count	Single-class Collapse	Consistency Reading
Track 1	QASMBench circuit execution under Aer fake backend	Q0/Q1/Q3	0	No	Public reproducible simulator/noise observer proof
Track 2	Quantinuum QV public circuit-trial result record	Q0/Q1/Q3/Q4	0	No	Real public QPU execution-record observer proof
Track 3	Google QEC decoder-prediction record	Q0/Q1/Q2/Q3/Q4	0	No	Real public QEC/decoder observer proof

Report-ready Consistency Sentence

Track 1, Track 2, and Track 3 use different public quantum data layers, but each layer can be organized into Q-state review-reference outputs without source-system modification and without collapsing into a single class.

11. Observer-layer Alignment Feasibility

The experiment supports observer-layer feasibility within the defined public-reference scope. It shows that public quantum-computing records can be separated into stable reference, watch, review-candidate, execution-uncertainty, boundary, or unavailable layers without issuing quantum-performance judgments or source-system commands.

Alignment Interpretation
The protocol uses existing public quantum records and derived public aggregate fields to organize review-oriented observer states. This preserves the boundary between observation and source-system control, provider performance evaluation, or QEC/decoder improvement claims.

12. What This Experiment Shows / Scope Boundary

Shows	Scope Boundary
Public quantum records can be organized into observer-layer Q-state review outputs.	Quantum advantage remains outside scope.
Benchmark circuit, hardware-result, and QEC/decoder records can be handled as distinct observation units.	QPU control, circuit optimization, and provider comparison remain outside scope.
Result distribution, heavy-output, entropy, syndrome/correction, and metadata fields can form public feature groups.	Error-correction improvement and decoder improvement remain outside scope.
QX unavailable handling can remain separated from ordinary review states.	QX is not an anomaly or failure verdict.
The three-track design supports quantum-domain observer-layer feasibility.	Production deployment, operational approval, and private backend data remain outside scope.

12.1. Interpretation and Public Artifact Boundary

Track 1 should be read as a small-scale reproducible grammar validation rather than a statistical claim. Q-state outputs are review-priority coordinates derived from public-reference observation criteria, and their correspondence to ground-truth review priority or operational triage outcomes has not yet been established and remains a subject of future empirical evaluation. Public artifacts expose aggregate observer outputs while non-public record-level materials and implementation-sensitive observer materials are intentionally withheld. Track 3 concentration in Q3/Q4 should be interpreted as QEC-layer review sensitivity, not as QEC failure, decoder improvement, or hardware-performance judgment.

13. Next-step Recommendation

The recommended next step is to keep the Quantum Compute PoC as an independent public-reference layer. Track 1 remains the reproducible simulation/noise proof, Track 2 remains the public hardware-result proof, and Track 3 remains the high-complexity QEC/decoder proof. Additional quantum datasets may be reserved for follow-up work; they are not required for the first public reference report.

Layer	Dataset	Role	Status
Quantum Track 1	QASMBench + Qiskit Aer fake backend	Public reproducible observer proof	Complete canonical run
Quantum Track 2	Quantinuum Quantum Volume public data	Real public QPU execution-record observer proof	Complete canonical run
Quantum Track 3	Google Quantum AI QEC 6804040	Real public QEC / decoder / correction observer proof	Complete canonical run
Future optional layer	Additional Google QEC or other public quantum archives	Scale or stress-extension only	Not required for first public release

Appendix A. Metrics Dictionary

Metric	Value	Meaning
Track 1 selected circuits	10	QASMBench small circuit executions under Aer fake backend.
Track 1 Q0 / Q1 / Q3	2 / 2 / 6	Stable, noise watch, and execution uncertainty states.
Track 2 observation units	7,924	Quantinuum QV circuit-trial / raw result records.

Metric	Value	Meaning
Track 2 Q0 / Q1 / Q3 / Q4	2,445 / 2,748 / 2,330 / 401	Record-level Q-state distribution; QX=0.
Track 3 experiment directories	131	Google QEC surface-code experiment groups.
Track 3 observation units	522	Decoder-prediction observation records.
Track 3 Q0 / Q1 / Q2 / Q3 / Q4	42 / 65 / 67 / 258 / 90	QEC observer-state distribution; QX=0.
Single-class collapse	No across all tracks	Q-state grammar remains non-collapsed.

Appendix B. Source Notes and Raw Data Handling Boundary

Source	Public Handling	Boundary Reason
QASMBench	Public OpenQASM benchmark circuits downloaded from GitHub; selected small circuits used for Track 1.	QASM files used as public circuit source; not a provider performance dataset.
MQT Bench	Public benchmark tooling ZIP retained as extension source.	Not executed in the canonical Track 1 run.
Qiskit Aer / fake_manila	Public software/fake-backend simulation environment used for ideal/noisy counts.	No live QPU access or private backend data is required for Track 1.
Quantinuum QV	Public GitHub JSON archive with raw_results, heavy_outputs, heavy_ideal, and QV circuit fields.	Used only for observer-state review mapping, not QV superiority or provider comparison.
Google QEC Zenodo 6804040	Public dataset including executed circuits, collected samples, decoder corrections, and intermediate files.	Used only for public QEC execution-record observer feasibility, not decoder/QEC improvement.

Appendix C. Artifact Inventory

Package Layer	Included Public Artifacts	Public Use	Boundary
Full Report Document	PDF distributed separately from the public data ZIP	Public-reference reading and citation	No source-system control, provider comparison, or performance claim
Track 1 Aggregate Package	Circuit inventory summary, aggregate Q-state summary, public source inventory, README, manifest	Reproducible simulator/noise observer summary	Non-public record-level materials are outside the public package
Track 2 Aggregate Package	QV source and file inventory, aggregate Q-state summary, public evidence summary, README, manifest	Public QPU execution-record observer summary	No provider comparison, QV superiority, or record-level implementation disclosure
Track 3 Aggregate Package	QEC experiment/file summaries, decoder/correction aggregate summaries, aggregate Q-state summary, README, manifest	Public QEC/decoder observer summary	No decoder/QEC improvement claim or record-level implementation disclosure
Cross-track Summary	Track-level comparison tables, source registry, boundary note, checksum manifest	Observer-state comparison across heterogeneous quantum records	Percentages are within-track review-reference distributions

Final Public Boundary

Public observer-layer formulation and aggregate quantum PoC evidence only. Track 1-3 outputs are review-reference observer artifacts and do not constitute quantum advantage, provider comparison, QPU control, circuit optimization, decoder improvement, or error-correction improvement claims.